Major Advanced Topics

26433101

Ongonwou Fred

Phd Candidate
2343003

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Spin-Polarization Control of Photoelectrons Using Poincaré Fields

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Daniel Younis[1,*], Daniel F. Gordon[2], and Bahman Hafizi[2]

[1]Department of Physics and Astronomy, University of Rochester, Rochester, New York 14627, USA
[2]Naval Research Laboratory, Washington, DC 20375, USA

Spin-Polarization Control of Photoelectrons Using Poincaré Fields

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The lexicon

- What is a Poincaré field?
  - Nonstandard term refering to fields that comply to the symmetries of the Poincaré group
  - The Poincaré group is the group of isometries of the Minkowski spacetime (translations, rotations and boosts)
- The notation $\boldsymbol{A}_{jm}$, and subsequently (j, m), applied to the field thus gives some information about
  - the number of poles, specified by j
  - their spatial alignment in respect to the arbitrary z-axis m
- Spin-polarization refers to the bias in the orientation of spins in a beam of particle relative to a chosen direction
  - It's a statistical measure
  - not to be confused with the projection m_s of the spin of a particle, which is intrinsic

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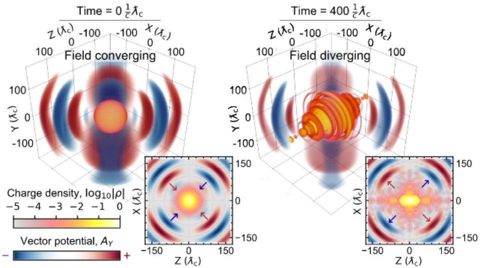
Electron charge density before and after the irradiation

- The target is Ne^{9+} $(Z = 10)$
- Natural units are used $(\hbar = c = 1)$
- Fig.1 Electron charge density before and after the irradiation by a 2-cycle magnetic quadrupole field



$$\gamma = \sqrt{2I_p/a_0^2 m_e} \ll 1, \qquad I_p/m_e = 1 - \sqrt{1-(Z\alpha)^2}, \tag{1}$$

$$\bar{\lambda}_c = m_e^{-1} \sim 3.86 \times 10^{-11}\,\mathrm{cm}, \qquad \tau_C = \bar{\lambda}_C/c, \tag{2}$$

$$a_0 = eE_0/\omega m_e \gg 1 \tag{3}$$

$$\gamma \sim 7.3 \times 10^{-3}, \qquad a_0 = 10, \qquad \omega = 0.07 m_e \sim 36\,\mathrm{keV}, \tag{4}$$

$$I_0 \sim 1.14 \times 10^{29}\,\mathrm{W/cm^2}, \qquad \lambda \sim 0.0347\,\mathrm{nm} \tag{5}$$

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Atomic and field potentials

- The system is described by the Dirac equation

$$\left(\gamma_\mu\left(\hat{p}^\mu + |e|\, A^\mu\right) - m_e\right)|\Psi\rangle = 0 \tag{6}$$

where $|\Psi\rangle$ is the bispinor wavefunction, γ^μ are the Dirac matrices, and p^μ and A^μ are the four-momentum and potential

$$A^0(\boldsymbol{r}) = V(r) = Z\alpha/\sqrt{|\boldsymbol{r}|^2 + \delta r^2}, \qquad \delta r \longrightarrow 0, \tag{7}$$

$$\boldsymbol{A}_{jm}(\boldsymbol{k}, t) = \mathrm{Re}\, A_0 \boldsymbol{\Phi}_{jm}\left(\theta_{\boldsymbol{k}}, \varphi_{\boldsymbol{k}}\right)\delta(|\boldsymbol{k}| - \omega)e^{i\omega t}, \tag{8}$$

$$\text{where} \qquad \boldsymbol{\Phi}_{jm}\left(\theta_{\hat{\boldsymbol{k}}}, \varphi_{\hat{\boldsymbol{k}}}\right) = \hat{\boldsymbol{k}} \times \boldsymbol{\nabla}_{\hat{\boldsymbol{k}}} Y_{jm}\left(\theta_{\hat{\boldsymbol{k}}}, \varphi_{\hat{\boldsymbol{k}}}\right), \qquad \hat{\boldsymbol{k}} = \boldsymbol{k}/\omega \tag{9}$$

$$j \geqslant 1, \qquad |m| < j - 1 \tag{10}$$

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Calculation of helicity

- The ion is initially in its ground state, written as $j = \ell + m_s = 1/2$
- The wavefunction $|\psi\rangle$ is obtained from (6) using a finite-difference scheme
- From the resulting wavefunction, we write

$$|\psi\rangle = |\psi_b\rangle + |\widetilde{\psi}\rangle \tag{11}$$

  - Bound part $|\psi_b\rangle$ is filtered through the masking

$$\mathcal{M}_\mu(\boldsymbol{r}) = 1 - \exp\left(-|\boldsymbol{r}|^4/\mu^4\right) \tag{12}$$

  - The resulting wavefunction is Fourier transformed
  - Low energy photoelectrons are then masked through $\mathcal{M}_\mu(\boldsymbol{p})$

- Helicity density is expressed as

$$\zeta(\boldsymbol{p}) = \langle\widetilde{\psi}(\boldsymbol{p})| \left(\boldsymbol{n}\cdot\frac{1}{2}\boldsymbol{\Sigma}\right) |\widetilde{\psi}(\boldsymbol{p})\rangle, \qquad \boldsymbol{\Sigma} = \boldsymbol{\sigma}\mathbb{1}, \qquad \boldsymbol{n} = \boldsymbol{p}/|\boldsymbol{p}| \tag{13}$$

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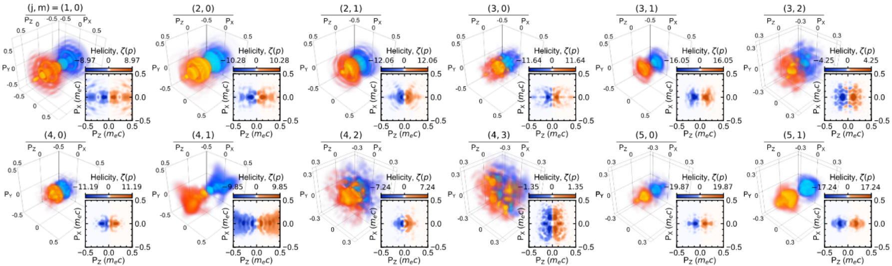
Helicity density $\zeta(\boldsymbol{p})$ of Ne^{9+} (Single mode ionization case)



- Fig.2 Momentum-helicity distribution $\zeta(\boldsymbol{p})$ for 12 individual cases
- The sign of p_z mostly decides the sign of $\zeta(\boldsymbol{p})$, despite $(3,2)$ and $(4,3)$
- More symmetries worth noting
  - $\int d^3\boldsymbol{p}\,\zeta(\boldsymbol{p})$ is negligible
  - For Dirac bispinor initialized in the e^- spin-down state, Fig. 2 will appear $\pi-$rotated through p_y
  - For irradiation modes with $m < 0$, helicity spirals are rotatting in the opposite direction

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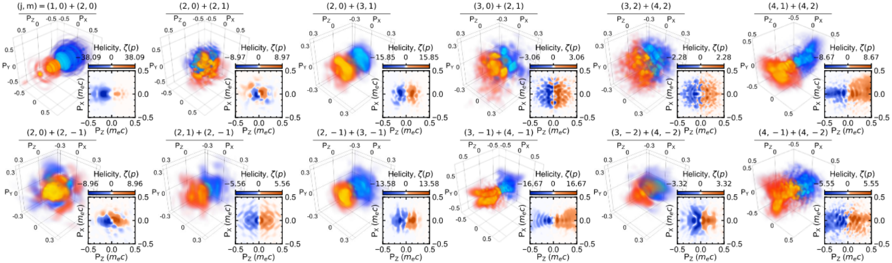
Helicity density $\zeta(\boldsymbol{p})$ of Ne^{9+} (Multimode ionization case)



- Fig.3 Momentum-helicity distribution $\zeta(\boldsymbol{p})$ for multipole mode combinations $(j_1, m_1) + (j_2, m_2)$
- Each combination has a unique photoelectron helicity signature. Which means that the helicity density may characterize the angular momentum content of a given radiation field
- Compared to the singlemode case
  - $\int d^3\boldsymbol{p}\,\zeta(\boldsymbol{p})$ not negligible anymore, notably $(1,0) + (2,0)$; $(4,1) + (4,2)$; $(3,-1) + (4,-1)$
  - $(3,-2) + (4,-2)$; $(3,2) + (4,2)$; $(4,-1) + (4,-2)$ present complex distributions with little regularity and high asymmetry

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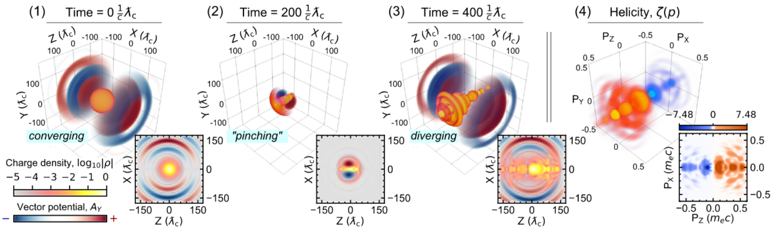
Effect of focal point misalignment



- Fig.4 Illustration of the effect of focal-point misalignment with the ion.
- Panels $(1)-(3)$: snapshots in time of an offset magnetic dipole $(1,0)$ irradiating Ne^{9+}
- Higher density photoelectrons in the $+z$-direction and suppression of negative helicity signal
- Sub-panels: cross-sectional views for y and $p_y = 0$ (panel 4)

It may be possible to exploit this misalignment in the focus to obtain relative control over the directionality, structure and degree of photoelectron polarization

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Conclusion

- Distribution of polarization states and angular momentum density of Poincaré fields can be used to produce structured electron helicity distribution
- This has been demonstrated numerically using an *ab initio* approach based on the time dependent Dirac equation
- This *ab initio* approach supersedes what is possible through S-matrix calculations, which are limited without an analytical solution to the Dirac equation